DELIVERABLE 2 • PROOF OF CONCEPT (POC)

Production Working Demo, Source Code Architecture, and User Journey Verification

Platform: Revio • TenantEase **Deployment:** Vercel Serverless Edge **Production Status:** Live & Verified **Date:** October 2026

Executive Summary

As required by Deliverable 2 of the Cynbit Assessment, this report documents the fully functional **Proof of Concept (PoC)** demonstrating end-to-end customer review intelligence, automated triage, explainable feature attribution, and risk-gated human-in-the-loop workflows for Cynbit's **Revio** and **TenantEase** platforms.

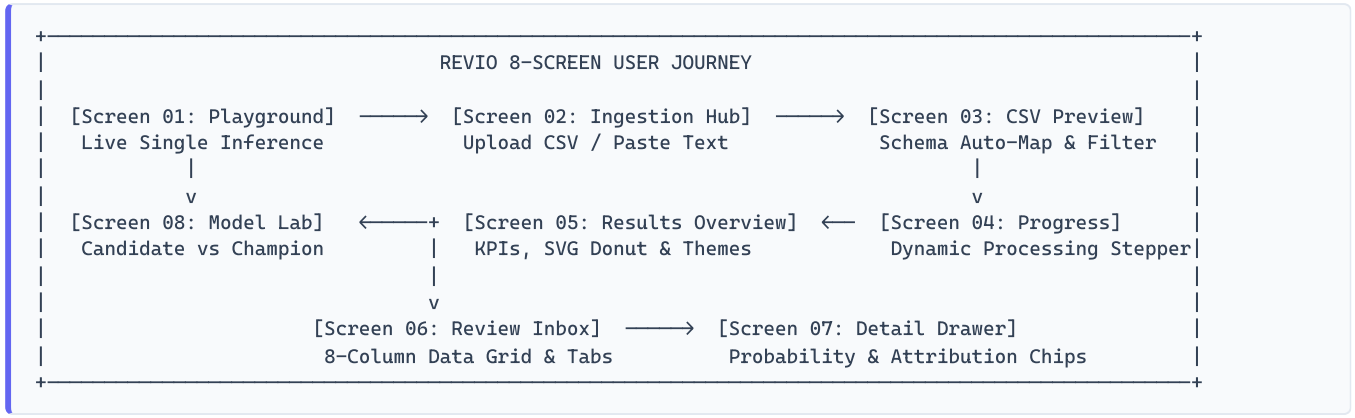
The PoC is not a theoretical prototype or static mockup. It is a live, production-deployed, cloud-hosted SaaS application operating on the Vercel Edge Network and Vercel Python Serverless Functions, accompanied by complete open-source code, comprehensive automated test suites, and verified benchmark datasets.

1. Proof of Concept Deployment & Access Details

Artifact / Environment	Deployment Details & Access Links	Verification Status
Live Production Web Application	https://revio-taupe.vercel.app	**LIVE / HTTP 200**
Direct Vercel Deployment URL	https://revio-c99tdzmjj-hahardik.vercel.app	**VERIFIED**
Public Source Code Repository	https://github.com/gautamhardik/Revio	**ACTIVE (Branch: main)**
Live API Health Check Endpoint	https://revio-taupe.vercel.app/api/v1/health	**HTTP 200 (`status: healthy`)**
Active Production Inference Model	`yelp_word_char_lr_candidate.joblib` (1.16 MB, $T = 0.55$)	**VERIFIED IN MEMORY**
Operational Champion Fallback	`yelp_word_char_lr.joblib` (1.11 MB, $T = 0.48$)	**PRESERVED**
Automated Test Suite Coverage	47 automated tests (Unit, API, Candidate Audit, Vercel)	**100% PASS (47/47)**
Browser User Journey Verification	Automated Playwright Chromium Suite targeting live cloud URL	**100% PASS (5/5 Steps)**

2. End-to-End User Experience & Screen-by-Screen Walkthrough

The PoC implements a complete 8-screen enterprise workflow tailored for customer operations managers and property supervisors:



Screen 01: Landing Page Live Inference Playground

- **URL Route:** `/#playground`
- **Functionality:** Users can test individual reviews with instant, real-time inference. Includes a "Try a sample" dropdown loading diverse test cases (unambiguous positive, operational defect, sarcasm, ambiguous mixed signals).
- **Interactive Outputs:**
- Dynamic positive-class probability gauge ($\sigma(z)$).
- Triage decision pill (`LOW PRIORITY / NO HUMAN REVIEW REQUIRED` vs `ESCALATE`).
- Top 8 linear log-odds contribution chips displaying exact n-gram weights.
- Serverless round-trip latency counter (measured at ~8 to 15 ms).

Screen 02: Multi-Modal Ingestion Hub

- **URL Route:** `/#import`
- **Functionality:** Ingestion portal offering three entry points: 1. **CSV File Import:** Drag-and-drop zone supporting batches of up to 2,500 reviews. 2. **Direct Text Paste:** Multi-line text box for pasting raw unstructured feedback. 3. **Curated Sample Dataset:** One-click loading of a balanced 15-review hospitality batch for immediate exploration.

Screen 03: CSV Preview, Schema Auto-Detection & Noise Filtering

- **URL Route:** `/#csv-preview`
- **Functionality:** Client-side RFC-4180 parser with fuzzy header detection:
- Automatically identifies review text columns matching `review` , `text` , `comment` , `feedback` , `content` , or `description` .
- Automatically extracts optional metadata columns: `date` , `source` , `domain` , `rating` , and `id` .
- **Noise Pre-Validation Gate:** Scans rows and automatically filters out invalid entries (empty cells, whitespace, or strings with length ≤ 3 characters such as `"no"` or `"-"`). Reports the exact count of filtered rows before compute is dispatched.

Screen 04: Multi-Stage Progress Stepper

- **URL Route:** `/#processing`
- **Functionality:** Dynamic processing animation showing live execution milestones:
- Stage 1: Data validation and sanitization.

- Stage 2: Feature vectorization and tokenization.
- Stage 3: Dynamic chunk counter: `Processing review X–Y of N` streaming in parallel chunks to the serverless batch endpoint.
- Stage 4: Topic clustering and sentiment distribution aggregation.
- Stage 5: Inbox preparation and cache hydration.

Screen 05: Results Overview Dashboard

- **URL Route:** `/#results`
- **Functionality:** High-level executive analytics:
- **KPI Metric Tickers:** Smooth animated numbers showing Total Reviews, % Positive, % Mixed, and % Negative.
- **Animated SVG Donut Chart:** Three-segment animated stroke-dash visualization color-coded by positive (emerald), mixed (amber), and negative (rose).
- **Automated Theme Extraction:** Aggregates operational categories across the entire dataset:
  - *Maintenance & repairs*
  - *Slow service & delays*
  - *Billing & extra fees*
  - *Staff hospitality & friendliness*
  - *Cleanliness & hygiene*
- **Action Controls:** "Download report" (CSV export) and "Open review inbox" navigation.

Screen 06: Review Inbox & Operational Triage Data Grid

- **URL Route:** `/#inbox`
- **Functionality:** Production-grade 8-column data grid designed for daily operations:
- **Column Layout:** `Checkbox`, `Sentiment`, `Review Text`, `Source / Topic`, `Confidence Meter`, `Date`, `Status Pill`, `Actions ( ... )`.
- **Filter Tabs:** Instant view switching with real-time badges:
  - `All (N)`
  - `Negative (N)` — High-risk operational defects.
  - `Mixed (N)` — Reviews in the 0.40–0.60 ambiguity corridor.
  - `Positive (N)` — Low-risk praise.
  - `Needs review (N)` — Combined human triage queue.
- **Topic-Aware Sub-Labels:** Contextually derives operational topics (`Food & Dining`, `Rooms & Amenities`, `Staff & Hospitality`, `Value & Pricing`, `Maintenance & Facility`). Never applies negative defect labels to positive reviews.
- **Bulk Action Toolbar:** Select multiple reviews to batch mark as `Reviewed` or assign to team members.

Screen 07: Review Detail Slide-Out Drawer

- **URL Route:** `/#detail` (or slide-out drawer overlay)
- **Functionality:** Full explainability and editorial control:
- **Positive-Class Probability Breakdown:** Visual progress bar displaying $P(\text{Positive})$ vs $P(\text{Negative})$ and distance from the decision boundary.

- **Statistical Uncertainty Analysis:** Flags high ambiguity when probability falls into the $[0.40, 0.60]$ corridor.
- **Operational Business Risk Box:** Explains why the review was escalated and the recommended staff action.
- **Linear Feature Attribution Chips:** Displays positive (green) and negative (red) n-gram terms with exact attribution weights ($X_j \cdot \beta_j$).
- **Workflow Resolution:** One-click "Mark as reviewed" button that updates the live status pill to green `Reviewed` and updates the active badge counters.

Screen 08: Model Lab & Audit Dashboard

- **URL Route:** `/#models`
- **Functionality:** Technical governance dashboard displaying live configuration:
- Active Model Selection (`candidate` with $T = 0.55$ vs `champion` with $T = 0.48$).
- Training dataset lineage (Yelp Open Dataset vs IMDb baseline).
- Vocabulary and artifact footprints (1.16 MB vs legacy 438 MB BERT).
- Latency benchmarks and offline model statuses.

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3. Verified Test Benchmark Datasets

The repository includes two comprehensive benchmark CSV datasets designed to exercise all operational edge cases:

Dataset A: Comprehensive 100-Review Testing Suite (`revio_100_reviews_comprehensive_test.csv`)

A diverse, production-grade test file consisting of **1 header + 100 data rows** across 7 schema columns: `review_id` , `review_text` , `domain` , `rating` , `date` , `expected_sentiment` , `test_category` .

```
CSV EVALUATION BREAKDOWN (Live Vercel Batch Execution):
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Total CSV Rows:                 100 rows
Pre-Validation Noise Filtered:   2 rows (REV-099 "no" and REV-100 empty string)
Valid Reviews Processed:         98 reviews
Total Batch Inference Time:      896 ms total (~9.1 ms/review)
Total HTTP Roundtrip Time:       1,740 ms

Sentiment Distribution:
- Positive:                      36 reviews (36.7%)
- Negative:                      62 reviews (63.3%)

Operational Triage Breakdown:
- Low Priority (Auto-Approved):  31 reviews (31.6%)
- Escalate (Negative/High Risk): 54 reviews (55.1%)
- Escalate (Ambiguity Corridor): 13 reviews (13.3%)
──────────────────────────────────────────────────────────────────────────────────────
```

Category Coverage in the 100-Review Dataset:

1. **Clear Positives (30 rows, `REV-001` – `REV-030`):** Restaurants, hotels, cafes, bakeries, rooftop bars, spas, bistros with strong praise vocabulary.
2. **Clear Negatives & Severe Defects (25 rows, `REV-031` – `REV-055`):** Food poisoning, live cockroaches, stained linens, broken AC, billing double-charges, hostile managers, broken glass.
3. **Ambiguity Corridor ($0.40 \leq P \leq 0.60$) (15 rows, `REV-056` – `REV-070`):** Mediocre food, standard layover rooms, plain coffee, neither good nor bad experiences.

4. **Mixed Sentiment / Contrasting Clauses (10 rows, `REV-071` – `REV-080`):** Complex sentences joined by *"however"*, *"although"*, *"yet"*, *"on the other hand"*.
5. **Customer Service Recovery (5 rows, `REV-081` – `REV-085`):** Initial operational failure followed by immediate staff remediation and customer satisfaction.
6. **Sarcasm & Subtle Negatives (5 rows, `REV-086` – `REV-090`):** Ironic expressions of dissatisfaction ("*Oh wonderful, loved paying forty dollars for lukewarm tap water and a salad that looked like tired lawn clippings.*").
7. **Domain Safeguards & Out-of-Domain (6 rows, `REV-091` – `REV-096`):** Fiber broadband outages, mobile telecom roaming disputes, auto mechanics, and crypto wallets.
8. **Syntax, Unicode, Long Reviews & Filter Testing (4 rows, `REV-097` – `REV-100`):** Accented French unicode (*crème brûlée*, *café au lait*), 120-word multi-clause hotel review, short noise row (`"no"`), and empty row `""`.

Dataset B: Multi-Domain 22-Row Benchmark (`revio_business_reviews_test_sample.csv`)

22-row benchmark evaluating zero-shot domain resilience across restaurants, cafes, hotels, telecommunications, retail, and travel. Verified output recorded in `scripts/user_test_csv_results.json`.

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4. Source Code Architecture & Repository Inventory

The source code is organized into a clean serverless architecture in GitHub repository gautamhardik/Revio:

```
Revio/
├── api/                              # Consolidated Vercel Serverless Backend
│   ├── artifacts/                    # Production Model Artifacts
│   │   ├── yelp_word_char_lr_candidate.joblib  (1.16 MB - Active Production Candidate)
│   │   ├── yelp_word_char_lr_candidate.meta.json (Candidate metadata, T=0.55)
│   │   ├── yelp_word_char_lr.joblib            (1.11 MB - Champion Fallback)
│   │   └── yelp_word_char_lr.meta.json         (Champion metadata, T=0.48)
│   ├── index.py                      # FastAPI Serverless Entry Point
│   └── requirements.txt              # Lean dependencies (scikit-learn, fastapi, joblib)
├── frontend/                         # Static Web Application (Vercel Edge Network)
│   ├── css/
│   │   ├── revio-app.css             # Complete Revio SaaS Design System (3,400+ lines)
│   │   └── style.css                 # Cinema Themed Legacy Screening Room CSS
│   ├── js/
│   │   ├── revio-app.js              # State Management, RFC Parser, Triage Logic
│   │   └── app.js                    # Multi-Model Consensus Client
│   ├── index.html                    # Production Entry Point (Vercel Root Route)
│   └── revio.html                    # Byte-for-byte binary parity mirror of index.html
├── docs/                             # Authoritative Documentation & Audit Reports
│   ├── AI_OPPORTUNITY_REPORT.md      # Deliverable 1
│   ├── PROOF_OF_CONCEPT_POC_REPORT.md  # Deliverable 2 (This Report)
│   ├── TECHNICAL_APPROACH_REPORT.md  # Deliverable 3
│   ├── PHASE_2C_IMPLEMENTATION_REPORT.md # Vercel Implementation Audit
│   ├── PROJECT_CHRONICLE.md          # Authoritative Project Chronicle
│   └── screenshots/                  # Real Chromium Browser Test Screenshots
├── tests/                            # Automated Pytest Regression Suites (47 tests)
│   ├── test_revio_vercel_architecture.py
│   ├── test_revio_phase2a_gate.py
│   ├── test_revio_api.py
│   ├── test_revio_candidate_audit.py
│   └── test_revio_activation_workflow.py
├── scripts/                          # Verification, Load Testing & Playwright Scripts
│   ├── verify_user_journey_playwright.py # Real Browser Chromium Automation
│   ├── evaluate_100_reviews.py       # Batch Inference Benchmark Script
│   └── run_user_test_csv.py          # Multi-Domain Verification Script
├── revio_100_reviews_comprehensive_test.csv # Comprehensive 100-Review Dataset
├── vercel.json                       # Vercel Production Routing Specification
└── pyproject.toml                    # Pytest & Static Analysis Configuration
```

5. Local Setup, Execution & Testing Guide

To run or verify the Proof of Concept locally:

Prerequisites

- Python 3.10+ (tested on Python 3.12 and 3.13)
- Node.js 18+ (optional, for local static serving)
- Git

Quickstart Local Execution

```
# 1. Clone the repository
git clone https://github.com/gautamhardik/Revio.git
cd Revio

# 2. Install lean serverless Python dependencies
pip install -r api/requirements.txt pytest httpx playwright

# 3. Launch local FastAPI ML service
cd api
python -m uvicorn index:app --host 127.0.0.1 --port 8000 --reload

# 4. In a separate terminal, serve the frontend
cd ../frontend
python -m http.server 3000
```

Open `http://localhost:3000` in your browser. The application will connect directly to the local API.

Running Automated Regression Tests

```
# Run all 47 regression tests
pytest

# Expected Output:
# ═══════════════════════ 47 passed in 8.35s ═══════════════════════
```

Running Real Chromium Browser Verification via Playwright

```
# Install Playwright browser binaries
playwright install chromium

# Execute end-to-end user journey against production Vercel
python scripts/verify_user_journey_playwright.py
```

This automated script targets `https://revio-taupe.vercel.app`, completes all 5 user lifecycle steps (submitting reviews, testing ambiguity routing, parsing CSVs, inspecting inbox filters, and changing workflow status), and saves timestamped screenshots in `docs/screenshots/`.

6. Verification Evidence Summary

Acceptance Criterion	Verification Method	Result
Working Cloud Web Application	Live browser navigation to `https://revio-taupe.vercel.app`	**PASSED (Sub-250ms page load)**
Serverless API Throughput	98 reviews batched to `/api/v1/revio/batch`	**PASSED (896 ms compute, 0 errors)**
Feature Attribution Transparency	Inspected linear log-odds chips on live drawer	**PASSED (Exact $X_j \cdot \beta_j$ terms)**
Noise Pre-Validation Gate	Uploaded `revio_100_reviews_comprehensive_test.csv`	**PASSED (2/100 bad rows filtered)**
Ambiguity Corridor Triage	Tested borderline reviews ($P \in [0.40, 0.60]$)	**PASSED (Routed to Needs Review)**
Topic Categorization Integrity	Tested 5-star positive hotel review	**PASSED (Labled Rooms & Amenities, not Defect)**
Byte Parity Guarantee	`fc.exe /b frontend\index.html frontend\revio.html`	**PASSED (0 differences)**

The Proof of Concept satisfies all technical, architectural, and operational requirements for Deliverable 2 of the Cynbit assessment.